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Neural Demodulation of Hardware-Captured PSK and Star-QAM Waveforms With a Bounded Input Context

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Dataset, trained models, and analysis code are available at https://github.com/not-your-pancake/Ai-demod and archived via Zenodo at https://doi.org/10.5281/zenodo.23509606.

ABSTRACT

Learned receivers are often evaluated on data whose transmitted sequence is short and periodic, which lets an expressive model predict a symbol from its neighbours instead of demodulating it. We bound this failure mode by construction. Each input window holds fewer neighbouring symbols than are needed to determine its label, which we verify for every transmitted word by enumerating the measured label sequences. We test it on 154,628 hardware-captured symbols from five trainer boards across BPSK, QPSK and star 8-QAM. Labels are measured from the waveform, not taken from the transmitter's configuration, and Gaussian noise is added in software. A 21,000-parameter convolutional receiver operates on symbol-synchronous passband windows and is given neither the carrier frequency nor the constellation. It comes close to a classical reference receiver that is given both: at the lowest signal-to-noise ratio its 8-QAM symbol error rate is 8.6

INDEX TERMS Neural receiver, demodulation, deep learning, physical layer, hardware dataset, maximal-length sequence, memorisation, symbol error rate, quadrature amplitude modulation.

I. INTRODUCTION

LEARNING-based receivers recover digital modulation from sampled waveforms without an explicitly specified signal model [1]–[3]. A recurring difficulty in evaluating such systems is that the transmitted data is frequently a short pseudorandom sequence repeated many times. When that is so, a model with sufficient temporal context can predict the next symbol from preceding symbols alone, without extracting any information from the waveform. The reported error rate then measures sequence memorisation rather than demodulation, and held-out evaluation does not detect this if the same short sequence appears in both partitions.

This paper bounds the failure mode by construction and then tests what the bound leaves open. For every transmitted word we compute, from the measured symbol-label sequence, the shortest neighbouring context that determines a symbol exactly. The receiver's input window is cut so that the neighbouring symbols it contains always fall short of that length. No window therefore determines its own label by lookup, whatever the network's capacity or training. The bound is a property of the data presented to the network rather than of its architecture, and it can be checked by enumeration. It is also narrower than an impossibility result. A short context can still be partly predictive, and the transmitter's pulse response leaves a weak trace of neighbouring symbols inside the window. We measure both, and we use held-out words and a corrected shuffled-label control to test whether the receiver exploits them.

The contributions are as follows.

1) A hardware dataset of 43 dual-channel oscilloscope captures from five physically distinct trainer boards, spanning BPSK, QPSK and a two-ring star 8-QAM constellation, with a per-record verification protocol and full provenance (Section III).
2) A ground-truth procedure based on coherent averaging of the transmitted waveform. The recovered BPSK labels reproduce the transmitted bit sequence and the QPSK labels fit a single symbol mapping. The 8-QAM labels are verified as constellation-point classes only (Section IV).
3) A bounded-context input design. For every transmitted word, the neighbouring symbols inside a window are

too few to determine the window's label, which we verify by enumerating the measured label sequences. We also measure what the bound does not exclude: partial predictability of the context, and inter-symbol interference from the transmitter (Sections V-A, VI-B and VI-C).

4) A characterisation of the boards' own demodulators, including a lock threshold reproducing across all five units, and evidence that self-consistency is an invalid criterion for demodulator lock (Section VI-E).
5) Two negative results, reported because they bear directly on claims made elsewhere in this literature. A constellation-geometry loss term does not improve error rate, and the latent output it acts on collapses to a single point. A classical receiver's dependence on carrier knowledge does not constitute a practical disadvantage at any observation length tested (Sections VI-D and VIII).

We make no claim of improved error-rate performance. The learned receiver comes close to a classical reference receiver but does not match it at the lowest signal-to-noise ratios (Section VI-A). The contribution is that it gets this close without being given the carrier frequency, the constellation geometry or the decision boundaries. It still uses symbol-synchronous windows and record-level gain control, and its input cannot determine a label from context alone.

II. RELATED WORK

A. LEARNED PHYSICAL-LAYER PROCESSING

Learned physical-layer processing has been demonstrated for end-to-end autoencoder transceivers [1], for systems trained over the air [2], and for joint channel estimation and detection [3]. Modulation classification from sampled in-phase and quadrature data has received particular attention, together with the synthetic datasets used to benchmark it [4]. Broad reviews of the area are given in [5]–[7], and of distribution shift specifically in [8].

The useful axis for comparison is what each system is *given*. Table 1 records, for twenty-one recent learned receivers and demodulators, whether the carrier frequency, the symbol timing, the pulse shape and the constellation are supplied rather than recovered, together with the domain of the input and the origin of the evaluation data. The RML22 dataset paper [4] and the 6G vision article [5] are cited above but not tabulated, because neither evaluates a receiver of its own.

The distribution is uneven. Nine operate on baseband in-phase and quadrature samples and seven on frequency-domain subcarriers; four take symbols already at the output of a matched filter, and one operates on passband samples. Fifteen are given the carrier frequency and fourteen the symbol timing. Eighteen are given the constellation, and seventeen fix a known pulse-shape family without presenting it to the network. Nineteen are evaluated on simulated waveforms only. Of the two exceptions, Hanna *et al.* use hardware captures for validation [9], and Cammerer *et al.* also train their system over the air on software-defined radios [10].

A minority withhold more. Zhang *et al.* demodulate BPSK with a sliding-window one-dimensional convolutional network on raw passband samples, detecting phase-shift locations without explicit carrier synchronisation or symbol-period grouping [18]; that system is the closest to ours in input domain, and differs in being evaluated on simulated waveforms and in being given the constellation. Xiao *et al.* take raw baseband samples and compensate time and frequency offsets blindly while jointly predicting the modulation type and the symbol index [13]. Dakic *et al.* predict LoRa symbols directly from baseband samples and report robustness to emulated carrier and time offsets without explicit synchronisation modules [19]. Zhu *et al.* recover bit sequences from oversampled baseband sequences while handling symbol timing offsets blindly through a shifted binary cross-entropy loss [12]. Hanna *et al.* estimate carrier offset, timing and symbol rate blind and then decode with a minimum-distance receiver against the classified modulation's constellation [9], and are one of the two entries whose evaluation involves hardware.

The larger group operates in the frequency domain with synchronisation already established. DeepRx performs channel estimation, equalisation and soft demapping jointly and emits bit log-likelihood ratios for 5G [28], and its MIMO extension adds either a combining-based or a fully learned transformation layer [29]. Cammerer *et al.* train a bit-wise autoencoder that learns geometric constellation shaping and bit labelling jointly with iterative demapping and decoding [10], making it one of the three tabulated systems not given a constellation. A further group learns the demapping alone, from symbols already at the matched-filter output [20]–[23].

These systems are evaluated on held-out partitions. Eleven of the twenty-one report a random held-out split; the remaining ten do not state one. None reports a partition grouped by recording or by device, and none constrains the receiver's input context relative to the context complexity of the transmitted sequence, which is the property Section V-A imposes by construction.

B. EVALUATION VALIDITY AND MEMORISATION

This paper's premise is that held-out evaluation is insufficient when the transmitted sequence is short and periodic. Related concerns are long-standing in machine learning generally—train/test contamination, and models that exploit dataset artefacts rather than the intended signal—but the specific failure considered here is narrower: the test sequence itself is predictable from its own past, so a model with sufficient context can reach a low error rate without demodulating.

Work in radio-frequency sensing has begun to quantify the effect. Shulman shows that segment-level cross-validation causes severe data leakage and optimism in drone identification benchmarks, and that regrouping the evaluation by recording collapses the inflated accuracies [30]; among the studies surveyed here it is the only one that reports a recording-grouped split. The same failure is documented outside wireless: Brookshire *et al.* show that dividing recordings into segments without grouping by subject produces severe

TABLE 1. What recent learned receivers and demodulators are given. ***family*** denotes a fixed known pulse-shape family that is not itself presented to the network; ***blind*** denotes symbol timing recovered by the system; ***partial*** denotes a residual offset estimated from the signal; ***n.s.*** denotes a property the paper does not state, which is not the same as a property it does not use. ***Domains***: passband, baseband in-phase and quadrature, post-matched-filter symbols, frequency domain. ***Data***: simulated, hardware used for validation only, hardware also used for over-the-air training (***hw-train***), hardware as the primary source. Surveys, datasets and vision articles without an evaluated receiver are cited in the text and not tabulated.

System	Carrier	Timing	Pulse	Const.	Domain	Data
Zhu *et al.* [11]	yes	blind	family	no	baseband	sim.
Zhu *et al.* [12]	yes	blind	family	yes	baseband	sim.
Hanna *et al.* [9]	partial	blind	family	yes	baseband	hw-valid.
Xiao *et al.* [13]	no	blind	family	no	baseband	sim.
Zhu *et al.* [14]	yes	blind	family	yes	baseband	sim.
Chia *et al.* [15]	yes	yes	family	yes	baseband	sim.
Tonix-Gleason *et al.* [16]	yes	yes	family	yes	freq.	sim.
Liu *et al.* [17]	yes	yes	family	yes	freq.	sim.
Zhang *et al.* [18]	no	blind	family	yes	passband	sim.
Dakic *et al.* [19]	no	blind	family	yes	baseband	sim.
Zheng *et al.* [20]	no	yes	yes	yes	symbols	sim.
Allwin *et al.* [21]	yes	yes	yes	yes	symbols	sim.
Gansekoele *et al.* [22]	yes	yes	yes	yes	symbols	sim.
Polvani *et al.* [23]	yes	yes	yes	yes	symbols	sim.
Wang *et al.* [24]	yes	yes	family	yes	baseband	sim.
Chen *et al.* [25]	yes	yes	family	yes	freq.	sim.
Gorday *et al.* [26]	no	yes	family	yes	baseband	sim.
Fischer *et al.* [27]	yes	yes	family	yes	freq.	sim.
Honkala *et al.* [28]	yes	yes	family	yes	freq.	sim.
Korpi *et al.* [29]	yes	yes	family	yes	freq.	sim.
Cammerer *et al.* [10]	yes	yes	family	no	freq.	hw-train
This work	no	yes	family	no	passband	hw-primary

leakage and heavily inflated performance in translational electroencephalography studies [31].

The underlying concern is long-standing in machine learning generally. Carlini *et al.* give a quantitative framework for testing whether a sequence model has unintentionally memorised rare training sequences [32], and Arpit *et al.* show that gradient-based optimisation learns simple patterns before it memorises noise [33]. These results establish that the partition matters, and that memorisation is measurable after training. None of them limits, before training begins, what context a receiver can draw on. We are not aware of prior work in learned communications that constrains receiver context below the context complexity of the test sequence, and that absence is part of the motivation for Section V-A.

C. HARDWARE-CAPTURED RF DATASETS

Public RF datasets are commonly synthesised, or captured from a small number of transmitters with ground truth asserted from the transmitter's configuration rather than measured from the emitted waveform. The distinction matters here: Section III shows a case in which the configured data word is *not* a reliable statement of the transmitted symbol sequence, and Section IV obtains labels by measurement instead.

Table 2 lists twelve public radio-frequency datasets against three axes: the number of physical devices, the granularity of the labels, and whether the data are captured from hardware or simulated. Nine are captured primarily from hardware, one uses hardware for validation only, and two are simulated.

The axis that matters here is label granularity. Seven of the twelve label at device level and three at modulation-class level. The remaining two are radio-map datasets whose targets are path loss and time of arrival rather than properties of a received signal. None labels individual symbols. Hanna *et al.* release the largest, ten million packets from 174 transmitters recorded across 41 receivers over a month [35]; Elmaghbub *et al.* capture 25 LoRa transceivers across indoor, outdoor and wired testbeds [38]; Zahid *et al.* capture 27 radios [39]; Morin *et al.* twenty [40]; Kashani *et al.* thirteen [42]; and Jagannath *et al.* ten commercial chipsets [43]. Our five units are fewer than any of these, and far fewer than the 174 of [35].

The distinction we draw is therefore not scale but what a label denotes. The datasets above answer which device transmitted a signal, or which modulation it carries. A symbol-

TABLE 2. Public radio-frequency datasets. ***n.s.*** denotes a quantity the dataset does not state; ***n/a*** denotes a category that does not apply. The device counts are as reported. Ours is smaller than every stated count except the three of [34].

Dataset	Devices	Label	Source
Hanna *et al.* [35]	174	device	hw-primary
Vrieze *et al.* [36]	n.s.	mod.-class	hw-primary
Al-Shawabka *et al.* [37]	n.s.	device	hw-valid.
Elmaghbub *et al.* [38]	25	device	hw-primary
Zahid *et al.* [39]	27	device	hw-primary
Morin *et al.* [40]	20	device	hw-primary
Utrilla [41]	n.s.	mod.-class	hw-primary
Allahham *et al.* [34]	3	mod.-class	hw-primary
Kashani *et al.* [42]	13	device	hw-primary
Jagannath *et al.* [43]	10	device	hw-primary
Jaensch *et al.* [44]	n/a	radio map	sim.
Yapar *et al.* [45]	n/a	radio map	sim.
This work	5	symbol	hw-primary

TABLE 3. Acquisition and signal parameters. All values were measured from the captured records rather than taken from the equipment specification.

Parameter	Value
Sample rate	87,260 Sa/s
Samples per record	523,600
Record duration	6.0004 s
Channel 1	TP-20, modulated output
Channel 2	TP-9, recovered data
Trigger	external, from TP-1
Carrier	1199.91 Hz
Symbol rate	599.955 baud
Carrier cycles per symbol	2 (exact)
Samples per symbol	145.44
BPSK data source	on-board 63-bit m-sequence
QPSK, 8-QAM data source	24-bit configurable words

level label answers which constellation point was transmitted at a given instant, which is what a demodulator must predict, and which Section IV obtains by measurement from the waveform rather than by assertion from the transmitter's configuration.

D. CLASSICAL RECEIVER THEORY

Classical treatments of the receiver functions replaced by learned systems—carrier recovery, timing recovery, matched filtering and detection—are given in [46] and [47]. The timing-error detector referenced in Section VIII is due to [48]. Properties of maximal-length sequences, including the linear recursion used in Section V-A, are standard and are set out in [49].

III. EXPERIMENTAL SETUP AND DATASET

A. APPARATUS

Signals were captured from MCM31/EV communication trainer boards using a two-channel digital oscilloscope, shown in Fig. 1. Channel 1 observes TP-20, the modulated output, and channel 2 observes TP-9, the board's own recovered data. The two are acquired simultaneously so that network input and hardware baseline originate from a single record. Acquisition is externally triggered from the board's symbol clock at TP-1.

[TO WRITE — Fig. 1 board labels]
Both reviews note that Board-1 to Board-5 are not legible on the photograph. Add visible B1–B5 annotations to fig0_setup.png.

Acquisition parameters were fixed for every record and are given in Table 3. The time base was chosen so that each record contains at least 57 repetitions of the longest data word, which is what makes the coherent averaging of Section IV effective. A shorter time base was evaluated and rejected: it yields too few repetitions for a stable signal-to-noise estimate.

The carrier is coherent with the symbol clock at exactly two cycles per symbol. This is the property that permits coherent averaging: the passband waveform repeats exactly at the word period, so averaging over repetitions suppresses noise without distorting the signal.

TABLE 4. Per-record verification battery. Every record in the corpus passes every check except where noted in Section III-D.

Check No.	Check	Pass condition
1	Channel count	2 channels present
2	Sample count	520,000–526,000
3	Duration	5.99–6.01 s
4	Sample rate	87,000–87,500 Sa/s
5	Quantisation levels	≥ 90 distinct
6	Carrier (4th power)	1199.5–1200.5 Hz
7	Word-period repetition	$r \geq 0.95$
8	Coherent-template SNR	≥ 15 dB
9a	Same word, two records	similarity ≥ 0.99
9b	Constellation fingerprint	matches the expected word
9c	Different words	similarity < 0.99
10	TP-9 agreement	logged, not gating

B. UNITS UNDER TEST

Ten units were sourced. Five were functional; five were excluded following individual inspection, in each case because the on-board data generator had failed, either in the sequence-load latch or in the pseudorandom generator itself. On such a unit the modulator receives no data and TP-20 carries an unmodulated carrier: the coherent template contains all of its energy in the carrier bin and none in the sidebands, and no usable waveform exists to capture. All results below therefore rest on five boards, and leave-one-board-out is a five-fold evaluation.

C. PER-RECORD VERIFICATION

Every record was subjected to the battery of checks in Table 4 before admission to the corpus. The battery is not decorative: it rejected records during the campaign and each rejection corresponded to a real configuration fault.

Elaborating check No.(9a–9c) from Table 4, we see that check No. (9a) compares two records of the same configured word and detects a failure to reload the transmitter between captures. Check No. (9b) compares the multiset of per-constellation-point occupancy counts against the value predicted by the configured word; it is invariant to constellation rotation and to the mapping between transmitter configuration and transmitted symbol, and therefore identifies the loaded word without assuming that mapping. Check No. (9c) is the converse and is required for held-out words, each of which is captured only once: two *different* configured words must not produce the same waveform. Neither (9a) nor (9b) detects a stale word in that case, because words designed to cover the full constellation share a fingerprint by construction.

Check No. (9c) must be evaluated on the *signed* similarity. Two words that are bitwise complements produce waveforms that are exact negations of one another, and an unsigned similarity reports these as identical when they are merely antipodal.

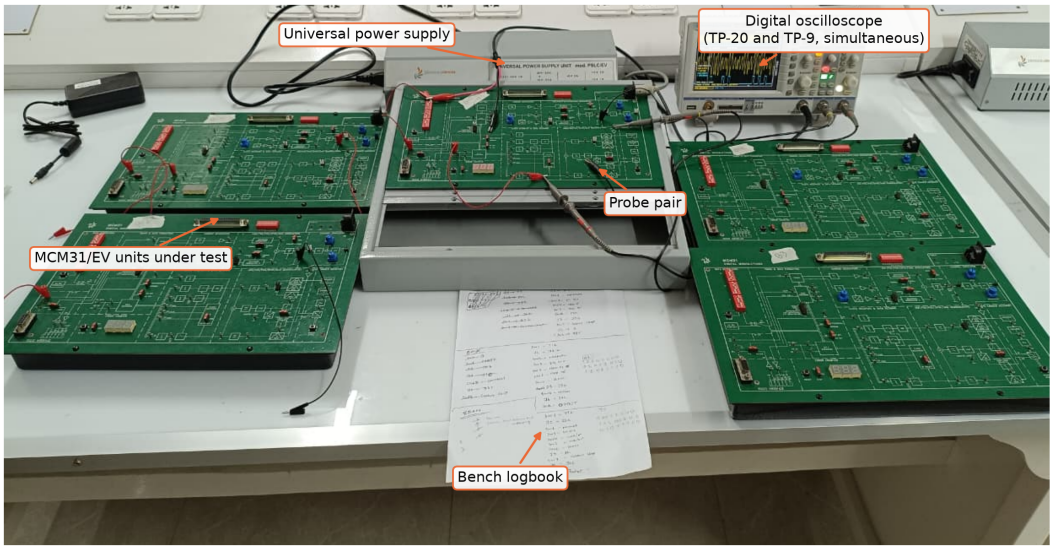


FIGURE 1. Measurement bench. The five MCM31/EV trainer units, labelled Board-1 to Board-5, correspond to boards B1–B5 in Table 10 and Fig. 8. One unit is under test at a time, powered from a shared supply, with the oscilloscope probing TP-20 and TP-9 simultaneously and triggered externally from TP-1.

TABLE 5. Dataset composition. *Words* counts distinct configured bit patterns; *independent* accounts for the antipodal equivalence described in the text.

Modulation	Records	Symbols	Words	Independent
BPSK	16	57,536	2	2
QPSK	12	43,152	3	3
8-QAM	15	53,940	6	5
Total	43	154,628		

D. CORPUS

The corpus comprises 43 records: fifteen from board B1 (seven main-campaign records and eight additional data words held out for evaluation) and seven from each of B2–B5. Segmentation at the symbol rate yields 154,628 labelled symbols, distributed as in Table 5. Coherent-template SNR across the accepted corpus is 22.7–43.2 dB, median 30.2 dB.

The design is not fully crossed. The primary words PRBS63, Q1 and T1 are recorded on all five boards: PRBS63 three times per board and Q1 and T1 twice. The additional words S1, Q2, Q3 and T2–T6 are recorded once each, on B1 only. Pattern identity and board identity are therefore confounded for those words. A leave-one-pattern-out fold that withholds one of them tests a new word on B1, not a new word on unseen hardware.

The distinction between *words* and *independent patterns* for 8-QAM is material. Two of the six words are bitwise complements of one another, and on this hardware complementary tribits map to antipodal constellation points: their coherent templates correlate at -0.998. After per-record amplitude normalisation the two carry identical information, and we therefore treat them as a single pattern throughout. Counting them separately would bias the leave-one-pattern-out result optimistically.

Four of the six 8-QAM words deliver six of the eight constellation points rather than the eight their bit patterns specify. The discriminator is a property computable from the bit pattern alone: a word is *framing robust* if all eight tribits survive every cyclic bit rotation. The two robust words deliver eight points; the four fragile words deliver six. This is consistent with the transmitter grouping bits into tribits from a fixed offset that does not coincide with the configuration order. It does not affect the labels, which are measured from the waveform, but it does mean the configured word is not a reliable statement of the transmitted symbol sequence which is precisely why the ground-truth procedure of Section IV does not use it.

E. MEASURED CONSTELLATIONS

Fig. 2 shows every symbol of every record, pooled across the five boards. The 8-QAM constellation is a four-phase, two-amplitude star of measured radius ratio 1.89 (range 1.867–1.914 over fifteen records). It is not a rectangular 16-QAM grid, and a loss function assuming rectangular geometry is inconsistent with this hardware.

IV. GROUND TRUTH

Labels are derived from the transmitted waveform, not from the transmitter's configuration. Fig. 3 shows the two processing paths and their separation.

A. COHERENT AVERAGING

Let $x[n]$ be the captured passband waveform and P the word period in samples, which is not in general an integer. The record is folded into K segments resampled onto a common grid of $M = \lfloor P \rceil$ points,

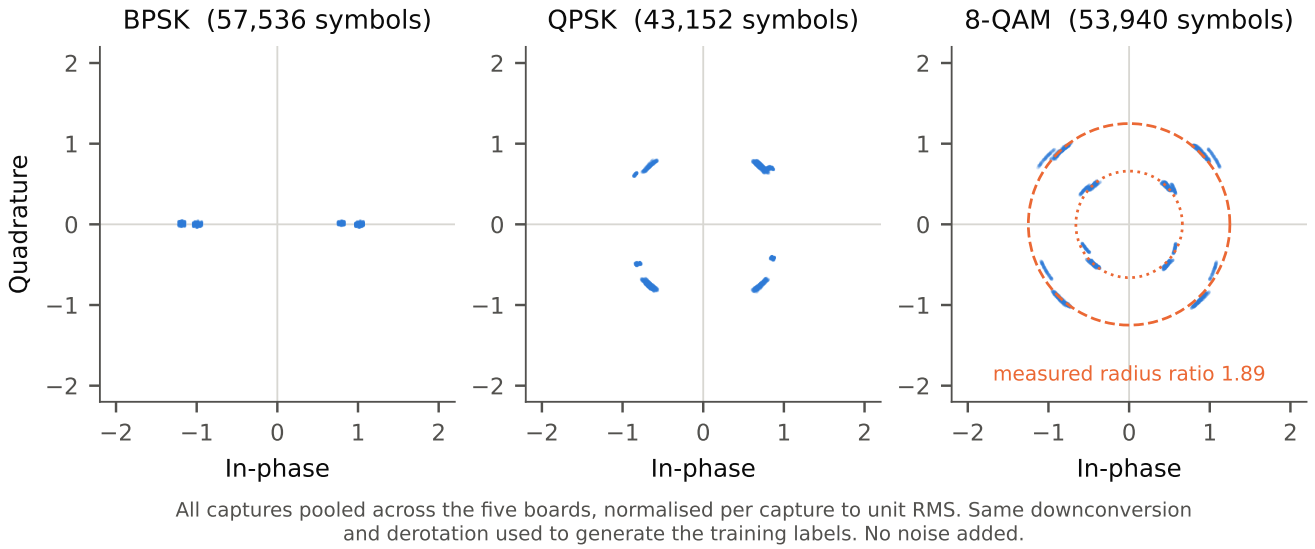


FIGURE 2. **Measured constellations: individual symbols, not the coherently averaged template. Each symbol is downconverted and integrated over its own symbol interval. Each record is derotated by its own whole-record *m*th-power estimate (7) and normalised to unit RMS. All records are pooled across the five boards, and 6,000 symbols per panel are drawn at random. The spread therefore shows per-symbol capture variability, including device-to-device differences. The 8-QAM panel shows the two rings at the measured radius ratio of 1.89. No noise has been added.**

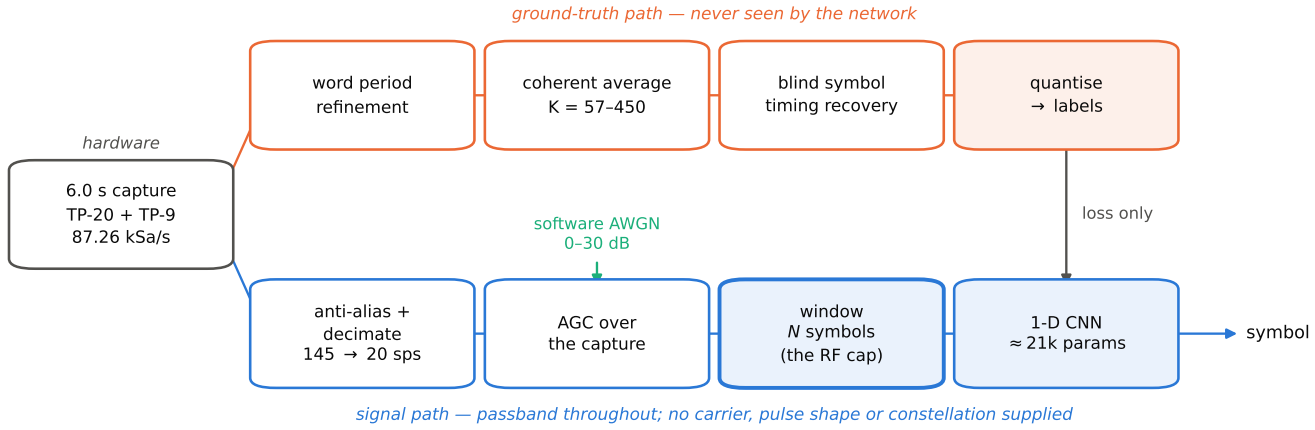


FIGURE 3. **Processing chain. The upper path produces ground-truth labels by coherent averaging; the lower path produces the network's input. The paths share the raw capture, and also the word period and the symbol timing $\tau^\star$ estimated from the coherent template. The network's windows are cut on the symbol grid that $\tau^\star$ defines, a dependency not drawn in the diagram. The label path reads the same filtered, resampled waveform from which the windows are cut (Section IV-B). At training and test time, software noise is added to each window first, and the window is then divided by its record's noise-free RMS (Section V-B). Labels reach the network only through the loss.**

$$x_k[m;P] = x\left(kP + m\tfrac{P}{M}\right), \quad \begin{matrix} k = 0,\ldots,K-1, \\ m = 0,\ldots,M-1, \end{matrix} \tag{1}$$

and the template is their mean,

$$t[m;P] = \frac{1}{K}\sum_{k=0}^{K-1} x_k[m;P]. \tag{2}$$

The period is refined by maximising the energy of the coherent average,

$$P^\star = \arg\max_P \ \mathrm{var}_m\left(t[m;P]\right), \tag{3}$$

searched over ± 6 samples in steps of 0.02 samples with M held fixed so that candidates are scored on a common grid. Writing $\sigma_r^2 = \mathrm{var}(x_k[m] - t[m])$ for the residual variance, the

template signal-to-noise ratio is reported as

$$\mathrm{SNR}_t = 10\log_{10}\frac{\mathrm{var}(t)-\sigma_r^2/K}{\sigma_r^2}, \tag{4}$$

in which the second numerator term removes the averaging bias, so the quantity reported is the single-repetition signal-to-noise ratio rather than the flattered post-averaging one.

B. SYMBOL EXTRACTION

To prevent the final symbol from falling on the wrap boundary, the coherent template is first extended circularly prior to filtering. The subsequent downconversion is achieved without requiring an external carrier estimate; because the transmitted word contains exactly $2N_s$ carrier cycles for N_s symbols, the downconverted sequence is computed as:

$$b[n] = t_{\mathrm{ext}}[n]\, e^{-j2\pi(2N_s)n/M}, \tag{5}$$

where $b[n]$ represents the downconverted discrete-time signal sequence, $t_{ext}[n]$ is the circularly extended coherent template waveform, n is the discrete-time sample index,N_s denotes the total number of transmitted symbols within the data word, and M is the number of grid points per word established in Section IV-A. This formulation remains exact regardless of the underlying sample rate.

Following downconversion, the sequence $b[n]$ is passed through a one-symbol boxcar, the matched filter for a rectangular pulse. Symbols are then sampled at instants $(i+\tau)T_s$, where T_s is the symbol period. The fractional timing offset $\tau \in [0,1)$ is recovered without data aid, by maximising the mean symbol magnitude on a $1/256$-symbol grid refined to $1/4096$:

$$\tau^\star = \arg\max_{\tau} \frac{1}{N_s}\sum_i \left|s_i(\tau)\right|. \tag{6}$$

where $s_i(\tau)$ denotes the complex filter output for the i-th symbol evaluated at the candidate timing offset τ, and i represents the discrete symbol index. Symbol-timing recovery is a prerequisite rather than a refinement. Extracting symbols without Equation (6) places decision instants part-way through transitions and collapses individual constellation radii—in our data, two of the eight 8-QAM radii fell to 0.20 against a nominal 1.31. This artefact is visually indistinguishable from a template wrap-boundary error and must be excluded before that diagnosis is reached.

Once timing is established, the residual constellation rotation is removed by an mth-power estimate:

$$\rho = \frac{1}{m}\arg\left(\mathbb{E}[s^m]\, e^{-jm\phi_{\mathrm{ref}}}\right), \qquad s' = s\, e^{-j\rho}, \tag{7}$$

where ρ is the estimated residual rotation angle, m is the modulation-dependent exponent ($m = 2$ for BPSK and $m = 4$ otherwise), and ϕ_{ref} is the target reference phase for the constellation. The variable s represents the optimally sampled complex symbol (corresponding to $s_i(\tau^*)$), s' is the resulting derotated symbol, and $\mathbb{E}[\cdot]$ denotes the sample mean taken over all $\approx$ 3600 symbols of the capture record. The rotation ϕ_{ref} must be applied inside the argument as written: for a reference phase of $45°$, the fourth-power sum lies near $180°$, which sits on the branch cut of the argument function. Subtracting the phase afterwards would erroneously place records whose phase is physically identical $90°$ apart. Furthermore, the expectation is calculated over the entire record rather than a single word, because a single word covering the constellation unevenly introduces enough bias to select an incorrect quadrant. Either of these errors would relabel every symbol of the affected record while leaving all per-record diagnostics deceptively intact.

The template supplies only the word period and $\tau^\star$. The labels are computed from the whole record, on the filtered, resampled waveform of Section V-B, whose symbol grid is set by $\tau^\star$. Each symbol is downconverted at exactly two carrier cycles per symbol and integrated over its own interval. All symbols of the record are derotated by (7) and averaged over the K repetitions of the word. The averaged word is then quantised. BPSK classes are the sign of the real part. QPSK classes are the phase quadrant. 8-QAM classes are the phase quadrant plus one bit for whether the radius exceeds the word's median radius. The word labels are tiled over the record. This whole procedure is offline: it uses the complete record, and nothing like it is available to the network at inference.

Because ϕ_{ref} is fixed, the labelling convention is common to all records. A class index refers to the same constellation point in every record. The evidence is that all 148 pairs of records carrying the same word, including pairs from different boards and attenuations, have label sequences identical up to a cyclic shift (Section VI-C).

C. VERIFICATION OF THE LABELS

Two checks test the recovered labels against what was transmitted. They do not give the same confirmation for every modulation.

1) BPSK has sixteen records. The fifteen PRBS63 records reproduce the board's 63-bit maximal-length sequence at 100%, against a chance level of 66.6% (95th percentile 69.8%). The chance level is computed over 63 cyclic shifts and both polarities. The one record of the 24-bit word S1 reproduces S1 at 100% under the same search. The carrier is recovered by fourth-power estimation at 1199.915–1199.918 Hz across all five boards.
2) A single memoryless bijection from constellation point to dibit fits all three QPSK data words simultaneously at 100%—36 symbols explained by one four-point mapping. Across 20,000 label permutations preserving the per-symbol occupancy counts, a perfect fit never occurred, giving $p < 5\times10^{-5}$.

The second check also resolves, for BPSK and QPSK, the mapping between transmitter configuration and transmitted symbol. For 8-QAM it cannot: a word covering all eight points exactly once admits a perfect bijection for any rotation, so the test is vacuous for a single word and remains unresolved.

The verified claims therefore differ by modulation. For BPSK, the label sequences are confirmed as the transmitted bit sequences. For QPSK, they are consistent with a single memoryless dibit mapping. For 8-QAM, the labels are measured constellation-point classes. They are confirmed only by the occupancy fingerprint of check 9b and by the cross-record consistency above. Which tribit each class carries is not established. Every 8-QAM result in this paper concerns recovery of constellation-point classes, not of transmitted bits. The context analysis of Section V-A uses the measured class sequences, which is the target the network is trained on. It therefore does not depend on the unresolved tribit mapping or on the framing offset.

The classical reference receiver of Section VI-A is not independent of this procedure. It uses the same downconversion, integration, derotation and decision regions, but on single noisy symbols rather than on a word averaged over 57–450 repetitions. Its agreement with the labels at high signal-to-noise ratio is therefore expected by construction. It is a reference for the error rate the labels permit, not an independent check of the labels.

V. RECEIVER

A. THE BOUNDED INPUT CONTEXT

Window geometry. The network receives a window of N consecutive symbols and labels the symbol at a fixed position within it: $[t-2, t-1, t, t+1, t+2]$ for BPSK, $[t-1, t]$ for QPSK and $[t]$ for 8-QAM, where t is the target. The *neighbouring* content of a window is therefore four symbols for BPSK (two on each side), one preceding symbol for QPSK, and none for 8-QAM. The target's own waveform is the signal to be demodulated and is not counted as context.

Context complexity. Let $y_0, \ldots, y_{L-1}$ be the cyclic sequence of measured labels of one word, with indices taken modulo L. Its one-sided context complexity κ^- is the smallest k for which y_i is a function of $(y_{i-k}, \ldots, y_{i-1})$ at every position i. The two-sided complexity $\kappa^\pm$ is the smallest k for which y_i is a function of the $\lfloor k/2 \rfloor$ preceding and $k - \lfloor k/2 \rfloor$ following labels. If a window's neighbouring content is shorter than this, some context in the word occurs with two different labels. No lookup, however learned, can then recover every label of the word from its neighbours. The quantities are computed on the measured label sequences, which are the targets the network is trained on, not on the configured bit strings. They are invariant to cyclic shift and to any fixed relabelling of the classes, so the unresolved 8-QAM tribit mapping does not affect them.

For PRBS63 the property also follows from the sequence's linear recursion. The bit string as listed for the board is the complement of the maximal-length sequence that satisfies

$$s[i+6] = s[i] \oplus s[i+5], \tag{8}$$

where $s[i]$ is the i-th bit of that sequence and $\oplus$ is modulo-2 addition [49]. Complementing every bit leaves every context property unchanged. Six consecutive bits determine the next one exactly; five do not. Thirty-one of the thirty-two five-bit patterns occur twice per period, once followed by each bit. The all-zero pattern occurs once and is followed by a one. Even a one-sided five-bit context therefore predicts the next bit at no more than 32 of 63 positions. The configured 24-bit words obey no such recursion: their minimum one-sided bit contexts are six to nine bits, and their symbol-level complexities are found by enumeration (Table 6).

TABLE 6. Context complexity of every measured word, in symbols. *Window* is the neighbouring context each window contains. *Lookup* is the accuracy of the best lookup from that context to the target label within the word, as if the neighbours were known without error. *Chance* is 1/(number of classes).

Mod.	Word	κ^-	$\kappa^\pm$	Window	Lookup	Chance
BPSK	PRBS63	6	10	2+2	0.508	0.500
BPSK	S1	7	7	2+2	0.625	0.500
QPSK	Q1	3	4	1+0	0.417	0.250
QPSK	Q2	3	3	1+0	0.583	0.250
QPSK	Q3	2	3	1+0	0.333	0.250
8-QAM	T1+T2	1	1	0	0.125	0.125
8-QAM	T3–T6	2	2	0	0.250	0.125

Every window falls short of its word's complexity, and the margins explain the caps. A third QPSK symbol would supply two preceding symbols, enough to determine every label of Q3. A second 8-QAM symbol would supply one, enough for T1 and T2, in which each of the eight points occurs exactly once per word. The BPSK cap of five symbols is conservative.

What the bound does not exclude. It excludes one thing exactly: deterministic lookup of a label from the neighbouring symbols in its window. It does not make the context uninformative. The *Lookup* column exceeds chance for S1 and for all three QPSK words, so a context cue could still act as a prior at low signal-to-noise ratio. Section VI-C measures that cue on the actual splits.

The bound also counts symbols, not signal dependence. Two mechanisms carry neighbouring-symbol information into a window. The first is the digital front end. The zero-phase anti-alias filter of Section V-B has a two-sided support of 257 raw samples, ± 0.88 symbol. A window sample can therefore depend on raw samples outside its own symbol interval. The filter's response is concentrated, though: only -50 dB of its energy lies beyond ± 0.5 symbol. The second mechanism is the transmitter's own pulse response, which no front end removes. We measure it on the word-averaged, noise-free waveform. For each symbol, we take the deviation of its 20 samples from the mean of all symbols of the same class; only its neighbours can cause that deviation. Relative to the symbol's own RMS, the deviation is -18.5 to -18.9 dB for BPSK, -21.0 to -25.2 dB for QPSK, and -24.2 to -27.9 dB for the 8-QAM words in which a point repeats. In units of half the minimum distance between class means, it is 0.12, 0.08–0.16 and 0.17–0.21. A one-symbol 8-QAM window therefore carries a weak trace of its neighbours. The bound applies to the symbols explicitly present in the digital window, not to every transmitted bit that can influence its

TABLE 7. Window geometry, receptive field and model size. *Context* is the number of neighbouring symbols before and after the target.

Modulation	Symbols	Context	Window	RF	Parameters
BPSK	5	2+2	100	93	20,932
QPSK	2	1+0	40	39	22,022
8-QAM	1	0	20	20	20,106

samples.

That trace has a consequence that the comparison of splits is designed to test. For T1 one preceding symbol determines the label, so on a split whose test records carry a training word, a receiver able to read the previous symbol through the trace could look its label up. On leave-one-pattern-out it could not, because a lookup learned on the training words is at chance on the withheld word (Section VI-C).

Enforcement. The bound is enforced through the *input* rather than through the layer stack. The network cannot attend to a symbol that is absent from its input, so the property is unaffected by later changes to the architecture. Table 7 gives the configuration.

For a stack of one-dimensional convolutions, the receptive field *RF* in input samples is calculated as:

$$\mathrm{RF} = 1 + \sum_{l} (k_l - 1) \prod_{j<l} s_j, \tag{9}$$

where k_l is the kernel size of the l-th convolutional layer, s_j represents the stride of the j-th layer, and the product is taken over all layers j preceding layer l. The sum includes the full-width combiner of Section V-C. The calculated *RF* is asserted against the window width in code. For BPSK the striding leaves the last 7 of the 100 samples outside the receptive field.

B. SIGNAL CONDITIONING

Each record's channel-1 waveform is low-pass filtered with a 129-tap FIR designed by the window method (Hamming window, cut-off 4 kHz at the 87.26 kSa/s capture rate; SciPy *firwin*). It is applied forward and backward (SciPy *filtfilt*), which gives zero phase and an effective two-sided support of 257 raw samples, ± 0.88 symbol (Section V-A). The filtered record is resampled by linear interpolation onto a grid of exactly 20 samples per symbol, with sample m of symbol i at raw position $(i+\tau^\star-\frac{1}{2})P_s+(m+\frac{1}{2})P_s/20$. Here $P_s \approx 145.44$ is the measured number of raw samples per symbol and $\tau^\star$ is the timing of (6). Symbols whose interval would extend beyond the record are dropped. The waveform is presented in *passband*. Conversion to baseband is not performed, as it would amount to supplying the carrier frequency.

The window for target symbol t is cut from this grid at the position given in Section V-A. A target too close to a record edge to fill its window is dropped. This removes two symbols at each end of every BPSK record and the first symbol of every QPSK record, so the evaluated BPSK and QPSK totals are 64 and 12 symbols below the corpus totals of Table 5. No 8-QAM symbol is dropped.

Amplitude is normalised by an automatic gain control computed over the whole record:

$$\tilde{x}[n] = x[n] \Big/ \sqrt{\tfrac{1}{L}\textstyle\sum_l x[l]^2}, \tag{10}$$

where $\tilde{x}[n]$ is the amplitude-normalised waveform sample at discrete-time index n, $x[n]$ is the filtered and decimated passband waveform sample, L is the total number of samples in the entire capture record, and l is the discrete summation index. This record-level normalisation removes device gain—one board's output is 40% below the others—and attenuator setting, while preserving symbol-to-symbol amplitude ratios. Per-window normalisation must not be used. An 8-QAM window is a single symbol, and forcing it to unit energy erases the inner/outer ring distinction, which carries one of the three bits per symbol. In our measurements, this held the 8-QAM symbol error rate at 0.50 with no improvement from 0 dB to noiseless, because the ring bit was pure guesswork.

The AGC is an offline, record-level quantity. Its denominator is computed from the complete *noise-free* record, before software noise is added. A deployed AGC would see the noisy signal, whose power exceeds the clean power by the factor $1 + 1/\mathrm{SNR}$. Our normalisation is therefore idealised, and the error rates reported here assume it.

Noise and the SNR axis. Additive white Gaussian noise is added in software to each window before (10) is applied, with per-sample standard deviation

$$\sigma = \sqrt{\tfrac{1}{L}\textstyle\sum_l x[l]^2} \;/\; \sqrt{10^{\mathrm{SNR}/10}}. \tag{11}$$

The SNR is therefore a per-sample power ratio at 20 samples per symbol. The noise is added after the anti-alias filter and occupies the full 0–6 kHz band of the decimated signal. For a real passband signal the corresponding symbol-energy ratio is $E_s/N_0 = \mathrm{SNR} \cdot 20/2$, that is $\mathrm{SNR} + 10$ dB. At the nominal 0 dB point, BPSK therefore operates at $E_s/N_0 = 10$ dB, which is why its error rate there is already of order 10^{-5}. The captures' own noise is not removed. Their template SNR of at least 22.7 dB puts it more than 13 dB below the added noise at every SNR up to 9 dB. The noise power is referenced to the record level, not to the window, so that all symbols in a record experience the same noise power. Referencing to the individual window would give inner-ring symbols less noise than outer-ring symbols, which no physical channel does. During training the SNR is drawn uniformly from 0 to 30 dB for each window, and a fresh noise realisation is drawn every time a window is used.

C. ARCHITECTURE AND OBJECTIVE

The receiver is a four-layer one-dimensional convolutional network. Each layer has 32 output channels, kernel size 5, no padding and no dilation, and is followed by a ReLU. The strides are $(1, 1, 1, 1)$ for the 20-sample 8-QAM window, $(2, 1, 1, 1)$ for the 40-sample QPSK window and $(2, 2, 2, 1)$ for the 100-sample BPSK window. These leave feature maps of length 4, 6 and 5. A full-width learned combiner takes the place of global pooling. It is a convolution with 32 channels

whose kernel spans the whole remaining length, followed by a ReLU, and it produces one 32-dimensional feature vector. Two linear output heads read that vector: symbol-class logits, and a latent $(\hat{I}, \hat{Q})$ pair. Global average pooling was deliberately avoided: it makes the effective receptive field equal to the whole input while a kernel-based calculation reports a smaller figure, so the number one would quote would not be the operative one. With these layers, (9) and the parameter counts of Table 7 follow directly.

The training objective is:

$$\mathcal{L} = \mathcal{L}_{\mathrm{CE}} + \lambda \min_{c \in \mathcal{C}} \left\| (\hat{I}, \hat{Q}) - c \right\|^2, \tag{12}$$

where $\mathcal{L}$ is the total training loss, $\mathcal{L}_{CE}$ is the cross-entropy loss derived from the symbol-class logits, and λ is a scalar hyperparameter determining the weight of the constellation-geometry penalty. The term $(\hat{I}, \hat{Q})$ represents the continuous two-dimensional coordinates produced by the latent output head, $\mathcal{C}$ is the measured constellation set from Fig. 2(comprising two antipodal points for BPSK, a single ring of four for QPSK, and a two-ring star of radius ratio 1.89 for 8-QAM), c is a candidate constellation point within $\mathcal{C}$, and $|| \cdot ||^2$ denotes the squared Euclidean distance.

Before the distance is taken, the latent outputs are scaled to unit RMS over each batch, and $\mathcal{C}$ is likewise normalised to unit RMS. The two heads share every layer up to the combiner output. The latent head receives gradient only from the penalty term, and the classification head only from $\mathcal{L}_{CE}$. The penalty is not tied to the true class: it pulls each latent output towards the *nearest* point of $\mathcal{C}$, whichever that is. It therefore supplies the shape of the constellation but not which point a given symbol should occupy, and a latent head that maps every input to one point incurs no penalty. Section VI-D examines what the head actually learns. All main results use $\lambda = 0.2$.

The cross-entropy term $\mathcal{L}_{CE}$ is class-weighted, with inverse-frequency weights computed from the training records of each fold, to compensate for the intrinsic imbalance of the transmitted words. Optimisation uses Adam at 2×10^{-3} with cosine decay over 25 epochs and a batch size of 512. These hyperparameters were fixed before the evaluation sweep and are identical in every fold. The 8-QAM radius ratio of 1.89 in $\mathcal{C}$ is the mean over all fifteen 8-QAM records, test records included. It is a single scalar, it enters only the penalty term, and Section VI-D shows that the penalty does not affect the error rate. The software was Python 3.14.5, NumPy 2.5.2, SciPy 1.18.0 and PyTorch 2.13.0 on a CPU.

D. EVALUATION SPLITS

Partitions are constructed at *record* level. Symbols within one record are repetitions of a single data word through a single signal path, so a symbol-level partition would place near-duplicates on both sides. Four splits are reported: leave-one-pattern-out, leave-one-board-out, leave-one-attenuation-out, and a random $80/20$ partition, the last included explicitly as the optimistic reference. Leave-one-pattern-out has 2, 3 and 5 folds for BPSK, QPSK and 8-QAM, leave-one-board-out 5 each, leave-one-attenuation-out 3, 2 and 2, and the random partition one fold each. That is thirty-five folds in total. T1 and T2 are withheld together as one pattern (Section III-D).

Every fold is trained once with seed 0. For the folds the comparisons rest on, two further seeds were trained: every leave-one-pattern-out fold, and every 8-QAM leave-one-board-out and random fold. The 8-QAM random partition was also redrawn with four further partition seeds. A reported mean over folds is the unweighted mean of the per-fold error rates unless stated otherwise. Error counts are given wherever a rounded rate could be mistaken for zero. Errors within one record are not independent, because every record repeats one word through one board. We therefore treat seeds and folds, not symbols, as the units of variability.

VI. RESULTS

A. ERROR RATE AGAINST A CLASSICAL REFERENCE RECEIVER

The reference receiver operates on the same filtered, resampled test record as the network, on the same symbol grid. It estimates the carrier from the fourth-power spectral peak of the whole noisy record, searched within ± 20 Hz of the nominal carrier. It then downconverts at that estimate and integrates over each symbol interval. This integrate-and-dump step is the matched filter for a rectangular pulse, not for the board's measured pulse. The residual rotation is removed by (7) over the whole record, and hard decisions are made on the decision regions of Section IV-B. The M-fold phase ambiguity left by (7) is resolved by choosing the rotation that agrees best with the true labels, a genie step the network does not get. Nothing is learned from training records.

Relative to the network, the reference receiver is therefore given four things: the constellation and its decision regions, a carrier search band, carrier and phase estimates from the whole record (about $3{,}600$ symbols, against the network's one to five), and the phase reference. Both receivers are given the symbol timing. Noise follows the same convention, (11), but not the same realisations. The reference receiver sees one continuous noise record per test record and SNR, while the network gets an independent draw per window. Table 8 and Fig. 4 report both on the same leave-one-pattern-out test records, aggregated the same way, as the unweighted mean over folds. Learned values are means over three training seeds, and classical values are means over five noise realisations.

At the lowest signal-to-noise ratios the reference receiver has the lower error rate for all three modulations. At 0 dB, 8-QAM gives 0.1310 ± 0.0008 (mean $\pm$ standard deviation over seeds) against 0.1206 ± 0.0016, which is 8.6% higher. QPSK gives 0.0091 ± 0.0012 against 0.0027, a factor of 3.4. Most of the QPSK difference comes from the Q1 fold, which trains on two records: 0.0175 against 0.0043 and 0.0054 for Q2 and Q3. BPSK makes 6, 19 and 30 errors in $57{,}472$ symbols across the three seeds, against about one for the reference receiver. From 6 dB, BPSK and QPSK differ by at most ten errors in

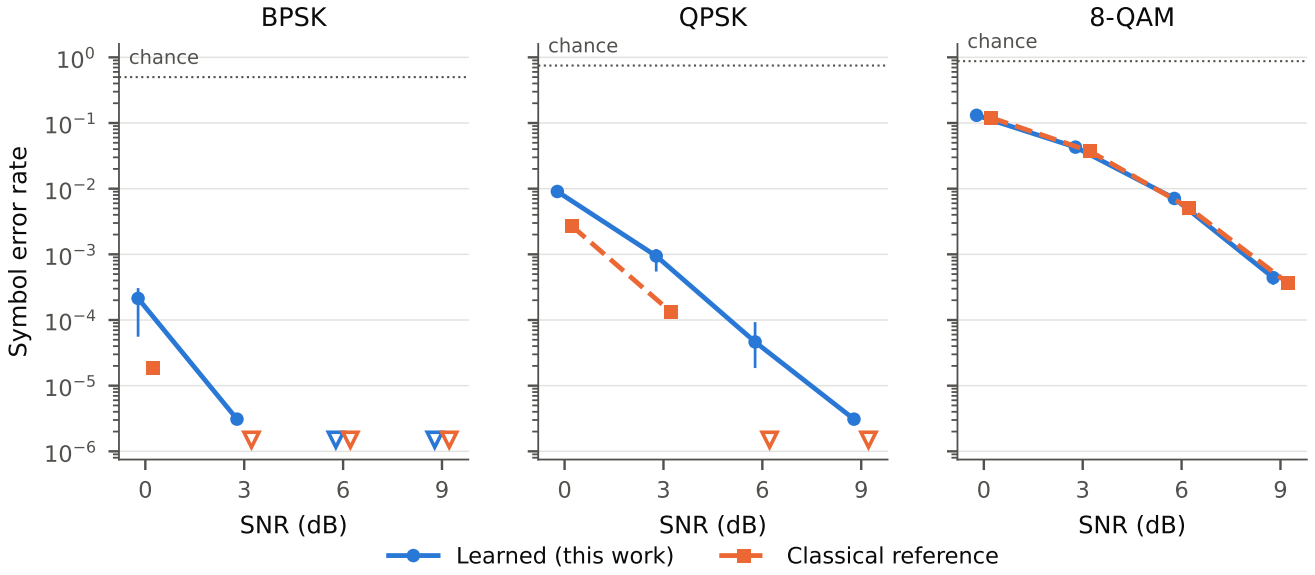


FIGURE 4. Symbol error rate against SNR for the learned and classical reference receivers on the same leave-one-pattern-out test records, as the unweighted mean over folds. SNR is the per-sample ratio of (11); E_s/N_0 = SNR + 10 dB. Learned: mean of three training seeds, with bars spanning the seeds. Classical: mean of five noise realisations. Hollow triangles mark operating points at which no error was observed in any run. They are drawn on a fixed floor line at 1.5×10^{-6}, a plotting convention below every non-zero mean, not a confidence bound.

TABLE 8. Symbol error rate, learned (L) and classical reference (C) receivers, on the same leave-one-pattern-out test records, as the unweighted mean over folds. L is the mean of three training seeds; C is the mean of five noise realisations. A dash means no error was observed in any run; $<10^{-4}$ marks rates that are non-zero but below the last digit shown.

	BPSK		QPSK		8-QAM	
SNR	L	C	L	C	L	C
0 dB	0.0002	$<10^{-4}$	0.0091	0.0027	0.1310	0.1206
3 dB	$<10^{-4}$	–	0.0009	0.0001	0.0429	0.0372
6 dB	–	–	$<10^{-4}$	–	0.0071	0.0051
9 dB	–	–	$<10^{-4}$	–	0.0004	0.0004
12 dB	–	–	–	–	$<10^{-4}$	0.0002

tens of thousands of symbols. For 8-QAM the gap narrows with SNR: 0.0071 against 0.0051 at 6 dB, and 0.00044 against 0.00037 at 9 dB. At 12 dB the learned receiver makes 5–18 8-QAM errors in 53,940 symbols, depending on the seed, and none from 15 dB. The reference receiver keeps a floor of nine errors per realisation at every SNR from 12 dB up. Where Table 8 shows a dash, no error was observed. That bounds the rate only loosely: errors within a record are not independent, so the usual $3/N$ bound (5×10^{-5} for BPSK) is indicative only.

These results support comparable performance in the tested regime. They do not establish equivalence, which we have not tested against any margin. A reference receiver given the constellation and whole-record synchronisation is close to the best a symbol-by-symbol detector can do on this channel, and we interpret neither difference as a performance claim.

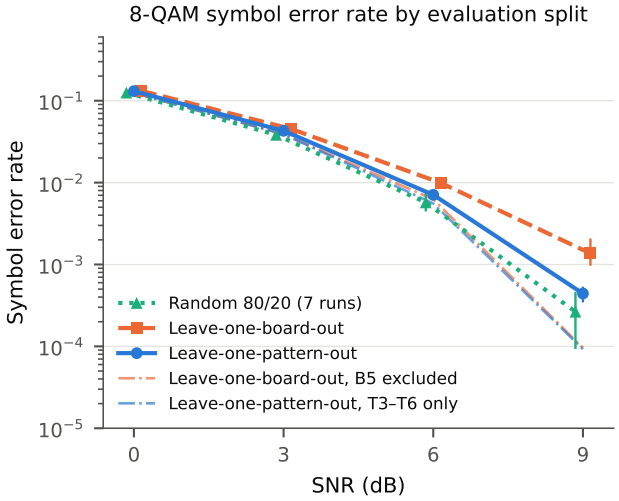


FIGURE 5. 8-QAM symbol error rate by evaluation split, as the unweighted mean over folds. Bars span three training seeds; for the random split they span seven runs (three seeds on the main partition and four further partitions). The dash-dot lines are leave-one-board-out without board B5 and leave-one-pattern-out without the T1+T2 fold, which trains on four records.

TABLE 9. 8-QAM symbol error rate by evaluation split, as the unweighted mean over folds, mean ± standard deviation over runs. ***Runs***: training seeds, or for the random split seeds and partitions. Leave-one-attenuation-out has one seed. T3–T6 are the leave-one-pattern-out folds that train on fourteen records.

Split (folds, runs)	0 dB	3 dB	6 dB	9 dB
Random 80/20 (1, 7)	0.1256±0.0022	0.0383±0.0028	0.0057±0.0010	0.0003±0.0001
Leave-one-pattern-out (5, 3)	0.1310±0.0008	0.0429±0.0007	0.0071±0.0005	0.0004±0.0001
T3–T6 only (4, 3)	0.1245±0.0024	0.0390±0.0007	0.0056±0.0006	0.0001±0.0002
Leave-one-board-out (5, 3)	0.1321±0.0007	0.0453±0.0017	0.0099±0.0007	0.0014±0.0006
without B5 (4, 3)	0.1241±0.0008	0.0388±0.0014	0.0063±0.0007	0.0001±0.0001
Leave-one-atten.-out (2, 1)	0.1367	0.0452	0.0071	0.0004

B. GENERALISATION

Withheld words. Averaged over all five folds, holding out a word costs 0.0054 at 0 dB: 0.1310 against 0.1256 for the random split. That is larger than the spread across random-split runs (0.0022). The whole difference comes from one fold. T1+T2 trains on only the four B1 records of T3–T6 and tests on all five boards, so it combines a withheld word with four unseen boards; it gives 0.1570 ± 0.0058. The four folds that withhold T3, T4, T5 or T6 train on fourteen records across all five boards. They give 0.1245 ± 0.0024, no worse than the random split at any SNR (Table 9). This is the comparison that Section V-A sets up. On the random split, the test records carry training words, so the neighbouring-symbol cues of Table 6 and Section VI-C are available, and for T1 they determine the label. On a withheld word they are not available. A receiver that exploited those cues would do better on the random split. It does not, within the run-to-run spread.

Withheld devices. Leave-one-board-out is worse than leave-one-pattern-out at 6 and 9 dB, but not uniformly across boards. Board B5 gives 0.0246 at 6 dB and 0.0064 at 9 dB (mean of three seeds), against 0.0057–0.0068 and at most 0.0002 for B1–B4. Without B5, the leave-one-board-out mean is 0.1241, 0.0063 and 0.0001 at 0, 6 and 9 dB, no worse than leave-one-pattern-out. One board in five does not transfer well. The rest transfer as well as a withheld word.

Per-fold results are given in Table 10. Two folds are thin and are reported with their training size: BPSK with the pseudorandom sequence held out trains on a single record and QPSK with its primary word held out trains on two. Both have the highest error rates in their modulation. Four of the five 8-QAM folds train on fourteen records. In those four folds, and in every BPSK and QPSK fold except Q1 and PRBS63, the test set is a single record from board B1 (Section III-D). They show generalisation to a new word on B1, not to a new word on unseen hardware.

C. NEGATIVE CONTROLS

Shuffled labels. The original control permuted labels across the training *and* test symbols of one leave-one-board-out fold. With the test labels randomised, a chance-level score was guaranteed whatever the network had learned, so the control could not fail. It has been replaced. Training labels are now permuted within the training records of each leave-one-pattern-out fold, the test labels are left as measured, and each fold is run with five permutation seeds. Averaged over folds and seeds, the shuffled-label error rates are 0.549, 0.710 and 0.922, against chance levels of 0.500, 0.750 and 0.875 (Table 11). Single runs scatter much more widely: from 0.169 to 1.000 for BPSK, 0.371 to 0.917 for QPSK and 0.625 to 1.000 for 8-QAM. This is a property of the data rather than of the network. Each word offers only 8 to 63 distinct noise-free windows. A network trained on permuted labels learns an arbitrary but systematic function of the waveform, often close to a constant class. Its error rate on a test word then depends on which classes that word contains: on an 8-QAM word with six of the eight points, a constant prediction scores 0.75 or 1.0. A BPSK run can even converge on an inverted demodulator. On this corpus, a single shuffled run is therefore not a reliable chance reference, and runs below chance arise by the same mechanism. What the control does establish is that no shuffled run comes near the trained receiver. On noise-free test windows the lowest shuffled-run error rate is 0.169. The receiver's is zero on every leave-one-pattern-out fold, and at most 0.0071 at 6 dB. A path from input to target that bypassed the waveform, such as an index, an ordering or a leaked label, would let a network trained on permuted labels still score well on the true test labels. None does.

Context-only. A third control measures the neighbouring context that the bound leaves in the window. A lookup from the true labels of a window's neighbours to its target label, fitted on the training records, gives 0.4959, 0.6943 and 0.9001 on the withheld words (Table 11), against chance levels of 0.500, 0.750 and 0.875. The cues of Table 6 do not transfer between words. On splits that reuse the training word they do: QPSK context then gives 0.583–0.596. For 8-QAM the window holds no neighbour. If the previous symbol could be read through the interference of Section V-A, however, a lookup from it would give 0.625 on the random split and 0.132 on leave-one-board-out (zero on four of the five boards), against 0.902 on leave-one-pattern-out. Section VI-B finds no corresponding advantage for the receiver on those splits.

Position-only. The position-only control requires a qualification that is easy to overlook. Labels are a periodic tiling of the transmitted word, so a symbol's index within that word determines its label exactly. Across all 148 pairs of records carrying the same word—including pairs drawn from different boards and different attenuations—the label sequences

TABLE 10. Per-fold results, training seed 0. *Train records* is the number of records in the training partition; the two folds with fewer than three are the thin folds discussed in the text. *Test symbols* counts evaluated windows, after the edge trimming of Section V-B. The S1, Q2, Q3 and T3–T6 folds each test a single record from board B1. *Position-only* is a lookup from a symbol's index within the data word to its most frequent training label, against chance levels of 0.500, 0.750 and 0.875. It is a meaningful control only where the withheld word is absent from training. Every leave-one-board-out fold reuses the training word, so position then determines the label exactly, and the entry is marked *n/a* (Section VI-C).

Modulation	Held out	Train records	Test symbols	SER at 0 dB	SER at 6 dB	Position-only
Leave-one-pattern-out						
BPSK	PRBS63	1	53,880	0.0001	0.0000	0.5290
BPSK	S1	15	3,592	0.0000	0.0000	0.5003
QPSK	Q1	2	35,950	0.0174	0.0001	0.7751
QPSK	Q2	11	3,595	0.0042	0.0000	0.7502
QPSK	Q3	11	3,595	0.0064	0.0000	0.8334
8-QAM	T1+T2	4	39,556	0.1519	0.0119	0.8864
8-QAM	T3	14	3,596	0.1310	0.0072	0.8749
8-QAM	T4	14	3,596	0.1313	0.0028	0.7500
8-QAM	T5	14	3,596	0.1271	0.0044	0.8751
8-QAM	T6	14	3,596	0.1185	0.0064	0.8751
Leave-one-board-out						
BPSK	B1	12	14,368	0.0001	0.0000	n/a
BPSK	B2	13	10,776	0.0000	0.0000	n/a
BPSK	B3	13	10,776	0.0002	0.0000	n/a
BPSK	B4	13	10,776	0.0000	0.0000	n/a
BPSK	B5	13	10,776	0.0002	0.0000	n/a
QPSK	B1	8	14,380	0.0070	0.0000	n/a
QPSK	B2	10	7,190	0.0054	0.0000	n/a
QPSK	B3	10	7,190	0.0049	0.0000	n/a
QPSK	B4	10	7,190	0.0036	0.0000	n/a
QPSK	B5	10	7,190	0.0174	0.0001	n/a
8-QAM	B1	8	25,172	0.1287	0.0057	n/a
8-QAM	B2	13	7,192	0.1194	0.0049	n/a
8-QAM	B3	13	7,192	0.1193	0.0057	n/a
8-QAM	B4	13	7,192	0.1283	0.0060	n/a
8-QAM	B5	13	7,192	0.1675	0.0302	n/a

TABLE 11. Controls on the leave-one-pattern-out folds, as the unweighted mean over folds. *Shuffled*: training labels permuted, test labels as measured; mean over five permutation seeds, with the range of the five fold-means in brackets. *Position-only*: the lookup of the text, with its cyclic-shift null mean in brackets. *Context-only*: a lookup from the true labels of the window's neighbouring symbols to the target label. Model: learned receiver at 6 dB, mean of three seeds. Chance is $1 - 1/K$.

Mod.	Chance	Shuffled	Position	Context	Model
BPSK	0.500	0.549 [0.376, 0.750]	0.5147 (0.4999)	0.4959	0
QPSK	0.750	0.710 [0.579, 0.846]	0.7862 (0.7477)	0.6943	$<10^{-4}$
8-QAM	0.875	0.922 [0.875, 0.977]	0.8523 (0.8719)	0.9001	0.0071

are identical up to a cyclic shift. A predictor reading only the symbol index therefore attains a symbol error rate of zero on any split whose test partition reuses a training word, which is the case for every leave-one-board-out, leave-one-attenuation-out and random fold. On those splits the position-only figure is not a control and we do not report it.

It is a control on leave-one-pattern-out, where the withheld word is absent from training and is not a cyclic shift of any training word: the best cyclic agreement between distinct words is 0.500 for QPSK and 0.375 for 8-QAM, against chance agreements of 0.250 and 0.125. There the lookup gives 0.5147, 0.7862 and 0.8523 (Table 11, Fig. 6). The nominal chance levels $1 - 1/K$ are not the right reference for single folds, because a word has few symbols and an unequal class mix. We therefore form a null for each fold. The lookup's predicted word is applied at every one of its cyclic shifts, which keeps its class mix and destroys any alignment with the test word. The null means are 0.4999, 0.7477 and 0.8719. A lookup that had learned something transferable would place the true alignment at or near the best of its shifts. In nine of the ten folds, other shifts do better: for QPSK, 10, 7 and 8 of the 11 other shifts give a lower error rate than the observed 0.7751, 0.7502 and 0.8334. For T4, the observed alignment is the best of the eight. An unaligned lookup lands there with probability $1/8$, which is unremarkable in one fold of ten. The position-only lookup therefore does no better than an unaligned copy of itself, which is the sense in which it is at chance.

This concerns the control, not the receiver. The network is given a window of waveform samples and nothing else: it receives no symbol index and classifies each window independently, so on no split can it exploit the positional redundancy that defeats the lookup.

D. CONSTELLATION GEOMETRY

The constellation term of (12) was ablated at $\lambda \in \{0, 0.2, 1.0\}$ (Table 12). The ablation is a separate, smaller run: one leave-one-board-out fold (B2 withheld), 18 epochs instead of 25, and one seed. Its absolute values therefore differ from the five-fold, 25-epoch means of Table 9. In the main run, the B2 fold gives 0.1194 and 0.0049 at 0 and 6 dB. Within the ablation, the three values of λ differ by at most 0.0030 at

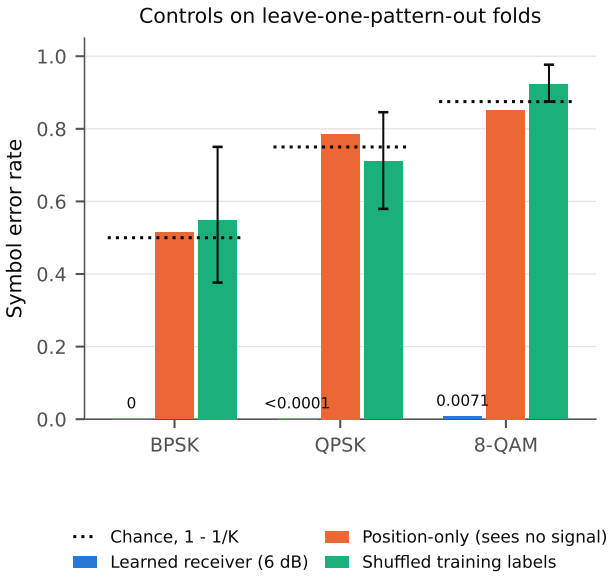


FIGURE 6. **Controls on the leave-one-pattern-out folds. Learned receiver at 6 dB, mean of three seeds. Position-only lookup. Shuffled training labels, mean of five permutation seeds, with bars spanning the five fold-means. Dotted lines mark the nominal chance level $1 - 1/K$; the spread of the shuffled control is discussed in the text.**

TABLE 12. **Ablation of the constellation term of (12). One leave-one-board-out fold (B2 withheld), 18 epochs, one seed; the main results use $\lambda = 0.2$ and 25 epochs.**

	8-QAM			QPSK
λ	0 dB	3 dB	6 dB	0 dB
0.0	0.1246	0.0370	0.0060	0.0058
0.2	0.1261	0.0359	0.0056	0.0051
1.0	0.1276	0.0364	0.0058	0.0054

0 dB, less than the spread across training seeds reported in Section VI-B. *The constellation prior does not improve error rate.*

Nor does the prior make the latent head reproduce the constellation. For 8-QAM, the latent outputs' mean distance to the nearest point of $\mathcal{C}$ is 0.250 at both $\lambda = 0.2$ and $\lambda = 1.0$. A random set of points of the same power gives 0.434. But 0.250 is exactly the value obtained when every output has the same magnitude: after normalisation such outputs lie on the unit circle, and the outer ring of $\mathcal{C}$ sits at radius 1.250. We checked the head directly. We retrained the ablation fold at $\lambda = 0.2$ and grouped the latent outputs of the noise-free test symbols by true class. The head has collapsed. For 8-QAM, all eight class means lie within 0.02 of one point, $(0.71, 0.71)$, and the spread within each class is at most 0.003. For QPSK, all four class means lie at $(-0.71, 0.71)$, the spread within each class is at most 0.005, and the mean distance 0.005 to the nearest point of $\mathcal{C}$ simply says that this single output sits on one constellation point. The head returns the same output for every input. That output satisfies the penalty while carrying no information about the symbol, exactly the degenerate solution noted in Section V-C. The constellation term therefore neither improves the error rate nor aligns the latent head with the constellation, and we do not present the head as evidence of recovered geometry. A penalty that tied each output to the point of its *true* class would avoid the collapse, but it would supply the constellation geometry as supervision.

The classifier's errors, by contrast, are geometrically structured. In the run of Fig. 7, 79.5% of the 863 8-QAM errors at 0 dB land on the nearest constellation point, against 14.3% if every wrong class were equally likely, an enrichment of $5.6\times$. For QPSK, all 33 errors do, against 66.7%. Of the 56 off-diagonal cells of the 8-QAM confusion matrix, 36 are exactly zero. The populated cells are the same-phase ring transition and the adjacent-quadrant transition, the two shortest distances in the constellation.

This is the pattern additive noise produces in any receiver that discriminates by waveform distance. It is consistent with the classifier having learned the constellation's geometry from data, although the loss never supplies class positions. It is not a test of memorisation. A model that had memorised waveform prototypes would confuse neighbouring prototypes under noise in the same way. The memorisation question is addressed by the bound of Section V-A and the controls of Section VI-C, not by Fig. 7.

E. EMPIRICAL PERFORMANCE OF THE ON-BOARD DEMODULATORS

The captures also characterise the boards' own demodulators, measured against the known transmitted sequence rather than by self-consistency. This measurement uses the fifteen BPSK PRBS63 records (Fig. 8). Agreement is the best agreement of the TP-9 output with PRBS63 over all 63 cyclic shifts and both polarities. Its chance reference, 66.6% with 95th percentile 69.8%, is the same maximisation applied to random sequences (Section IV-C). Across all five units, agreement is consistent with chance for all eleven records at or below 509 mV output level. Two of them exceed the 95th percentile of the null marginally, at 71.4% and 73.0%, about what eleven draws from the null would produce. Agreement reaches 92–100% for all four records at or above 626 mV. The interval between those two levels is the separation observed in our sampled records. No record falls inside it, so any threshold within it separates all five boards. A precise threshold is not established by these data.

Board-level behaviour differs in kind rather than degree. One board's BPSK path delivers 391 mV where the others deliver 626–663 mV and never reaches threshold, while its QPSK and 8-QAM outputs are nearly identical to the others (cross-board waveform correlation $+0.9987$ and $+0.9991$, against $+0.9769$ for BPSK). A second board crosses the threshold but locks imperfectly, at 92.1%.

We recommend against self-consistency as a lock criterion. A demodulator that has lost lock also repeats. On one board at 329 mV, self-consistency read 0.979 while agreement with the transmitted sequence was at chance: the unlocked demodulator had entered a stable period-two state, inverting

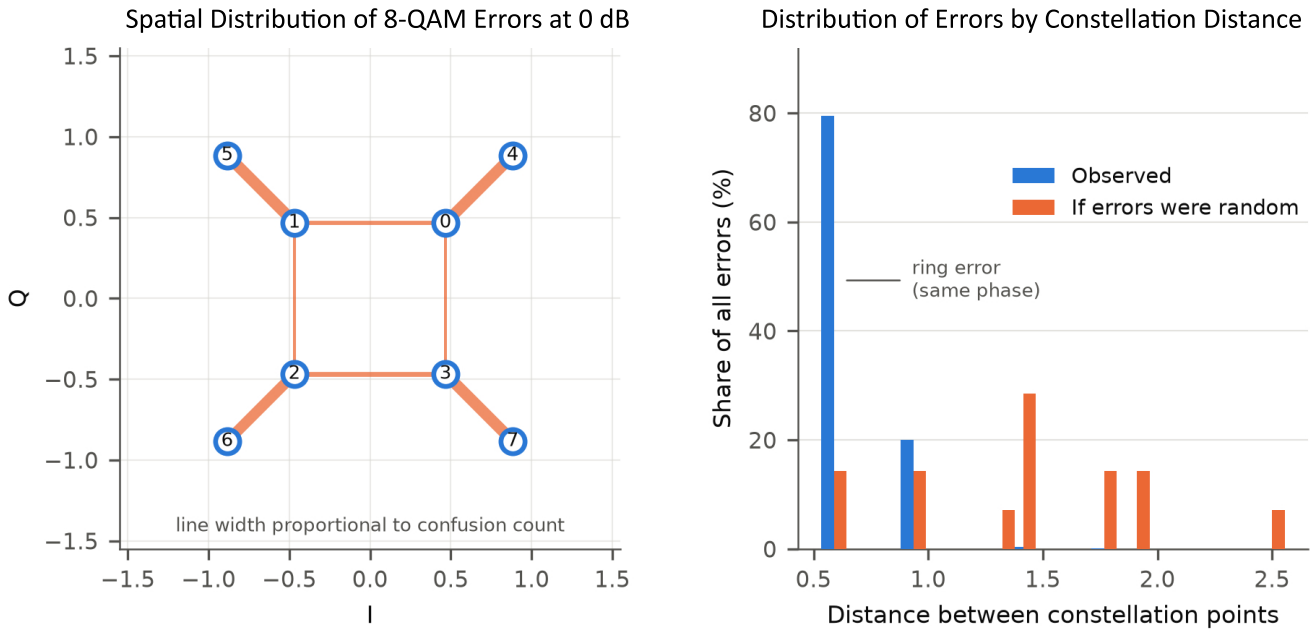


FIGURE 7. Destination of 8-QAM errors at 0 dB. The figure comes from a separate training run with board B2 withheld (leave-one-board-out, 18 epochs, $\lambda = 0.2$, seed 0), with one noise realisation: 863 errors among 7,192 test symbols. Left: confusions drawn on the unit-RMS reference constellation, with line width proportional to count. Each line combines both directions of a confusion; the two directions differ by 9.6% of all errors in total. Pairs with fewer than five confusions are not drawn. Right: share of errors against inter-point distance, compared with a null in which every wrong class is equally likely.

its output every word period (autocorrelation -0.93 at one word period, $+0.90$ at two). Four records in this study would have been misclassified as locked by that metric.

Finally, the boards' own QPSK demodulator failed to lock on the primary QPSK word on all five units, at both attenuator settings, although the fingerprint check confirms the word was correctly loaded. No hardware baseline is therefore available for QPSK.

VII. DISCUSSION

The principal result of this work is methodological. When a learned receiver is trained on a periodic transmitted sequence, held-out evaluation alone does not establish that the receiver demodulates. Part of the property can instead be imposed. If the input window is cut shorter than the context complexity of every transmitted word, no window determines its label by lookup, whatever the model's capacity or training procedure. That guarantee is narrow. It does not remove partial predictability of the context, and it does not remove the inter-symbol interference that carries a weak trace of neighbouring symbols into the window.

The empirical tests address what the bound leaves open. On splits that reuse a training word the neighbouring context is far more informative than on leave-one-pattern-out; for 8-QAM T1, one preceding symbol determines the label. A receiver that exploited the context through interference would therefore score better on the random split than on leave-one-pattern-out. Against the withheld-word folds with com-

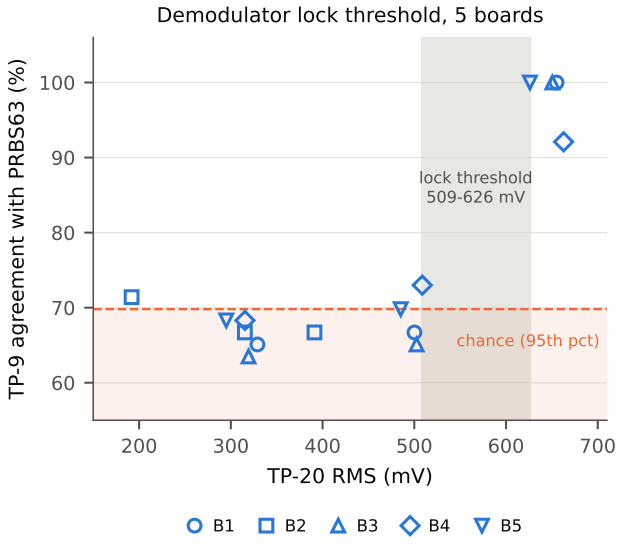


FIGURE 8. Behaviour of the boards' own BPSK demodulators on the PRBS63 word, across five boards: agreement of TP-9 with the transmitted sequence against TP-20 output level. The shaded band is the chance region up to the 95th percentile of the null (69.8%), from the best agreement over 63 cyclic shifts and both polarities. Records at or below 509 mV are consistent with chance; records at or above 626 mV agree at 92–100%. No record falls between the two levels.

parable training data it does not, beyond the spread across seeds and partitions. The one fold that is worse, T1+T2, trains on four records from a single board (Section VI-B). The corrected shuffled-label control adds a second, weaker check (Section VI-C). Neither test is a proof. Together with the bound, they make sequence exploitation an unlikely explanation of the error rates reported here.

We note that the bound is cheap to adopt. It requires the context complexity of the transmitted words, computable by enumeration from the measured labels, and an input window cut accordingly. Where a test sequence has high complexity the constraint is loose. Where the words are short, as in much laboratory practice, it is tight.

The comparison of Section VI-A should be read in this light. The classical reference receiver is given resources the network is not: the constellation's decision regions, a carrier search band, carrier and phase estimates from the whole record, and resolution of the phase ambiguity against the labels. It is also built from the same operations as the label procedure. It has the lower error rate at the lowest signal-to-noise ratios. The learned receiver comes close to it without being given the carrier frequency, the constellation geometry or the decision boundaries. It is not equivalent to it, and we make no equivalence claim.

The cross-device gap is narrower than it first appears. Leave-one-board-out is worse than leave-one-pattern-out at 6 and 9 dB because of one board, B5. Without B5, the leave-one-board-out mean is no worse than leave-one-pattern-out (Section VI-B). One board in five transfers poorly. That makes device-invariant training worth pursuing, but five boards cannot show how common such a board is.

Beyond the evaluation of the learned receiver itself, the empirical performance of the hardware baseline in Section VI-E is of independent interest to anyone evaluating aged laboratory equipment. That a demodulator which has lost lock can present a perfectly self-consistent output is not, in our experience, widely appreciated, and any protocol treating repeatability as evidence of correct operation is exposed to it. The failure is also invisible on an oscilloscope: an unlocked digital output looks entirely normal at any time base. We therefore recommend that lock be established only by comparison against a known transmitted sequence.

VIII. LIMITATIONS

Content diversity. The corpus is large in noise realisations and hardware variation but small in distinct transmitted content: 8-QAM comprises five independent 24-bit words, or 40 distinct symbol instances.

Device count and design. Five functional units support a five-fold cross-device evaluation. The device-to-device differences we observe are large enough to be measured, but the sample is too small to characterise their distribution. The design is not crossed: the additional words S1, Q2, Q3 and T2–T6 exist only on board B1, so those leave-one-pattern-out folds test new words on one board, and pattern and device effects cannot be separated for them. Recording every word on every board is the remedy.

Scope of the bound. The bound excludes deterministic lookup of a label from the neighbouring symbols in its window. It does not exclude partial predictability (Table 6) or the neighbouring-symbol trace left by the transmitter's pulse response, measured at -18.5 to -27.9 dB (Section V-A). That these are not exploited is an empirical finding of Section VI-B, not a consequence of the bound.

Offline, record-level processing. The labels, the symbol timing and the AGC are each computed from the complete record. The AGC uses the record's noise-free RMS (Section V-B). The network does not see the label path, but its windows inherit the record-level timing and gain. The receiver is therefore a classifier of pre-synchronised, pre-normalised windows, not a streaming receiver.

Run-to-run variation. Three training seeds were run on the folds the comparisons rest on, and a single seed elsewhere. With so few seeds and partitions, differences smaller than the seed spread reported in Section VI-B cannot be resolved.

Synthetic noise. The hardware supplies the carrier, the filter response, quantisation and device-to-device variation, while additive white Gaussian noise is applied in software. The boards' own noise control contributes approximately $+0.02$ dB within the signal band and is unusable as a signal-to-noise axis.

Symbol timing is supplied. The receiver is given neither the carrier frequency nor any description of the pulse shape. Its input window is symbol-synchronous, however, which supplies the symbol rate and the sampling phase, and the waveform it sees is shaped by the physical transmit pulse. This is not incidental: under a one-symbol window for 8-QAM, timing cannot be recovered from the window itself, so the context bound and timing-blindness cannot both hold. The system should be described as operating on symbol-synchronous windows from a conventional timing front-end [48].

Carrier sensitivity does not support a robustness claim. We measured whether a classical receiver's dependence on carrier knowledge constitutes a practical disadvantage. Estimating the carrier blind from eight symbols (13.3 ms) gives an error of 1.02 Hz, yet demodulating those same eight symbols with that estimate yields zero errors, because over 13.3 ms a 1 Hz offset corresponds to 4.9° of phase rotation. Estimate error and tolerance contract together, and the classical receiver was self-consistent at every observation length tested, from 8 to 3,596 symbols. Reporting a collapse under an externally imposed carrier offset would misrepresent a receiver that does not make that error, and we therefore do not report one.

8-QAM labels are classes, not bits. The tribit carried by each 8-QAM constellation point is not resolved (Section IV-C). The 8-QAM results concern constellation-point classification.

Operating regime. A single carrier frequency, a single symbol rate, and an additive Gaussian channel without multipath or fading. Generalisation beyond this regime is not evidenced here.

IX. CONCLUSION

We have presented a hardware dataset of 43 dual-channel captures from five distinct communication trainer boards, together with a convolutional receiver whose input window always holds fewer neighbouring symbols than are needed to determine its label, for every transmitted word in the corpus. The bound excludes deterministic lookup from the window's context by construction. The remaining routes, partial predictability and inter-symbol interference, are measured and tested rather than assumed away. The receiver comes close to a classical reference receiver on BPSK, QPSK and star 8-QAM without being given the carrier frequency, the constellation or the decision boundaries. It operates on symbol-synchronous windows with record-level gain control, under software-added Gaussian noise. Its errors concentrate on the nearest constellation neighbours. Two negative results are reported. A constellation-geometry loss term does not improve error rate, and its latent output collapses to a single point. A classical receiver's dependence on carrier knowledge did not, under measurement, constitute a practical disadvantage.

Future work should address the limitations that bound the present study most tightly: the number of distinct transmitted words, the restriction of the additional words to a single board, and the number of devices. These set the resolution of the leave-one-pattern-out and cross-device evaluations.

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[TO WRITE — AI-use disclosure]

IEEE requires AI-generated content (text, figures, code) to be disclosed here, naming the system, the sections affected and the level of assistance; language editing alone is outside the policy but disclosure is recommended. This revision was drafted with Claude (Anthropic), including substantive text in Sections I, II, IV–VIII. State the actual use of ChatGPT, Claude and Gemini accurately before submission.

[TO WRITE — prior dissemination]

Disclose the accepted WorldCUR 2027 abstract in the submission letter (and cite it here if it is archived), and state how this article extends it.

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